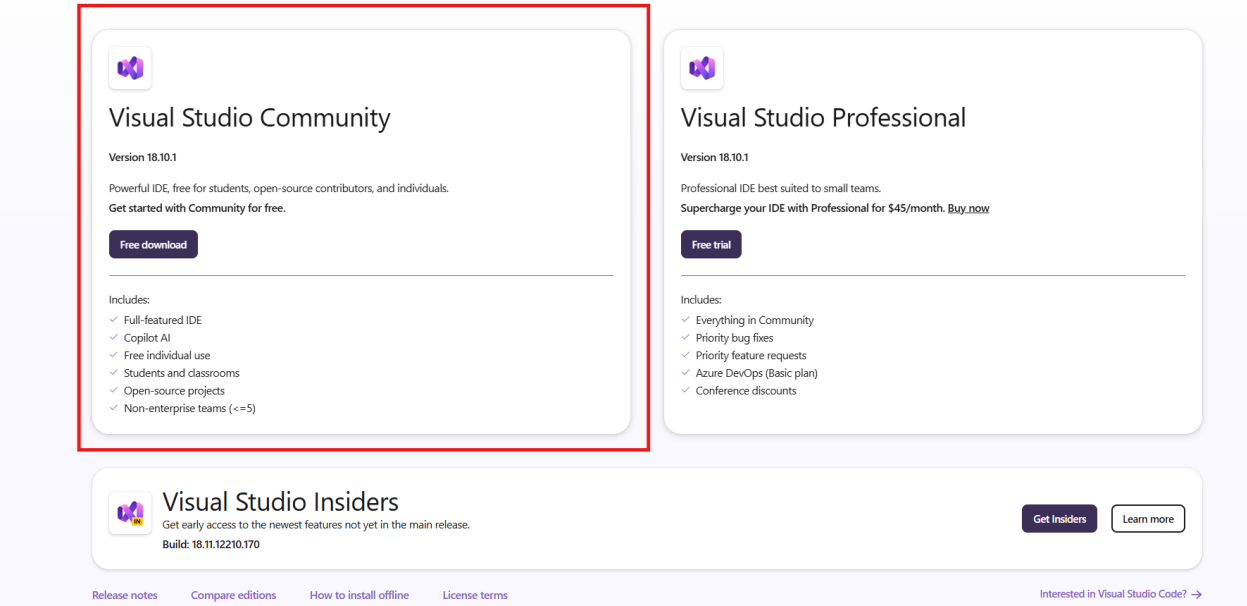
Set up Programming Environment

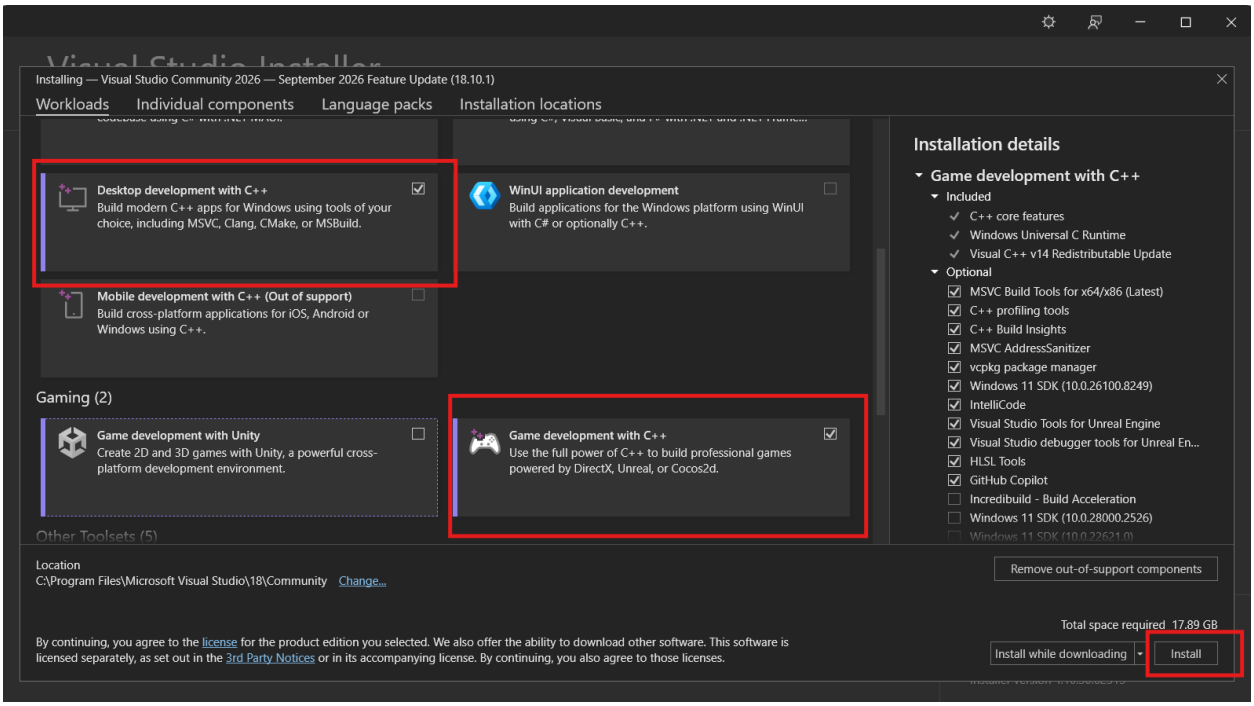
For this module, we will work with TL-Engine. This is an engine developed specifically for developing games at the University of Lancashire, and it aims to enable you to learn programming and different concepts of game development.

TL-Engine was developed and works only with Visual Studio. Hence, the first thing you need is Visual Studio on your computer. To download Visual Studio, visit the following link:

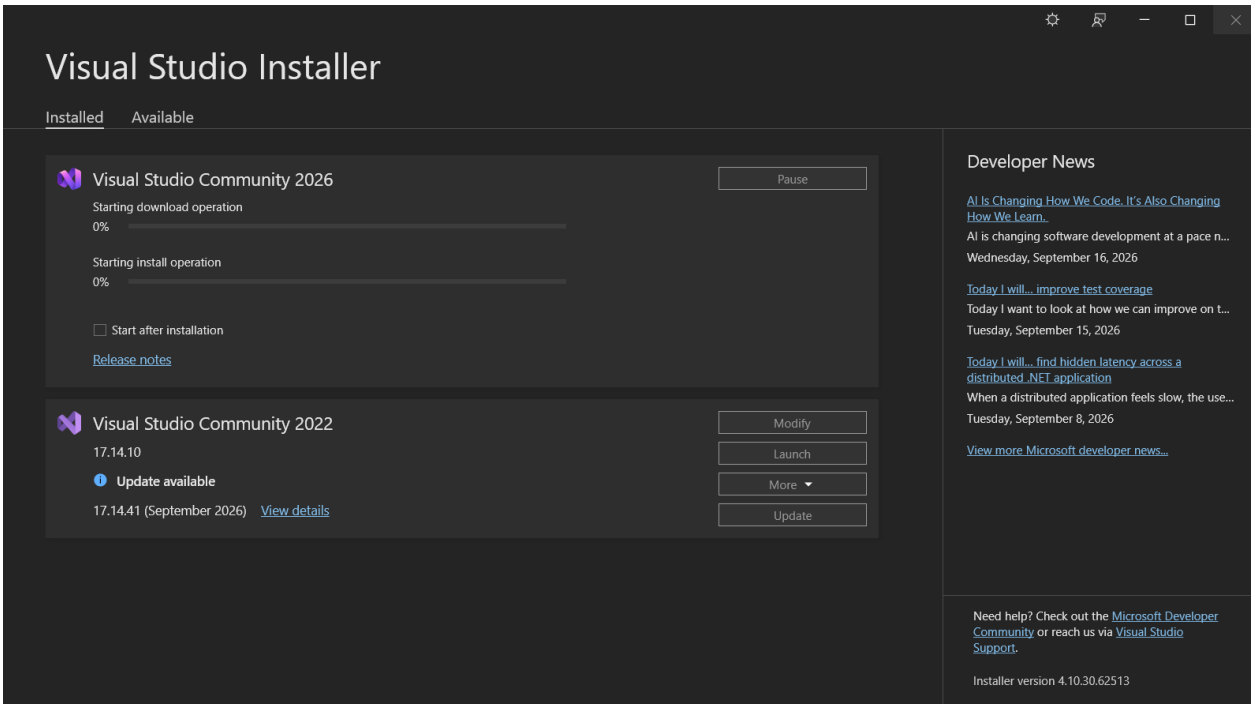
https://visualstudio.microsoft.com/downloads/



This link will open the window you see above, or something similar. Once you're there, select Visual Studio Community and download it. Once the installer is downloaded, run it. This will make a window similar to the one below open on your computer. Then, in the installer, select the Desktop development with C++ and the Game Development with C++ options, and click Install.



Pressing Install will make the window look like the picture below. Then wait until Visual is installed. Once it is installed, then press Finish and restart your computer.

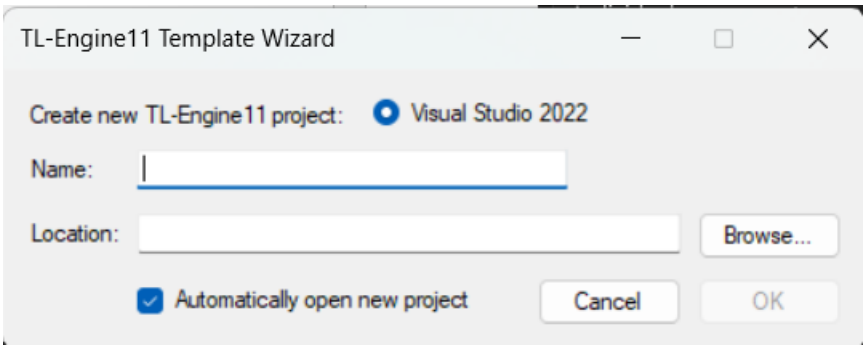


After installing Visual Studio, we need the actual engine. On Blackboard, find the installer and download it. Once the installer is downloaded, then open it and go through the installation process. This is a straightforward process.

Once TL-Engine is installed on your device, restart your computer so changes can be applied.

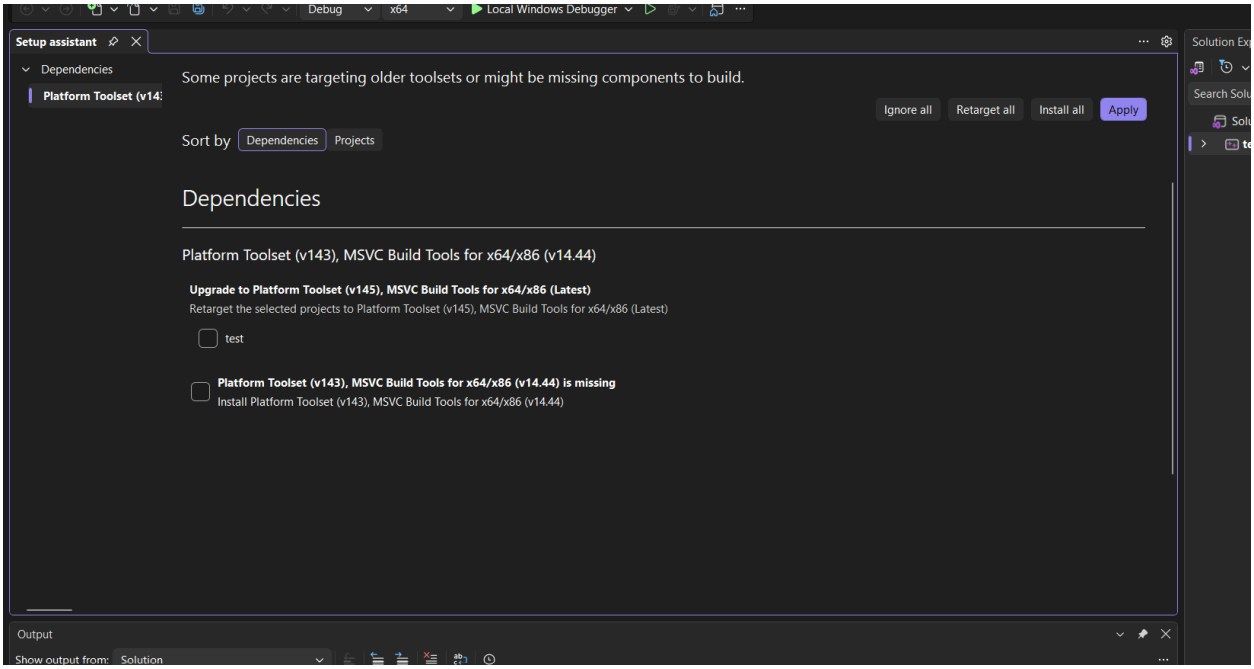
To test if TL-Engine is installed on your device, on the navigation bar of your computer, search for “Create new TL11 Project” and click on it. This will open a small window that looks like the figure below.

If on your computer there is no option “Create new TL11 Project”, search for TL-Engine11 or something similar. Open that folder and double-click on “TemplateWizard.exe”. This will open the small window like the one below.

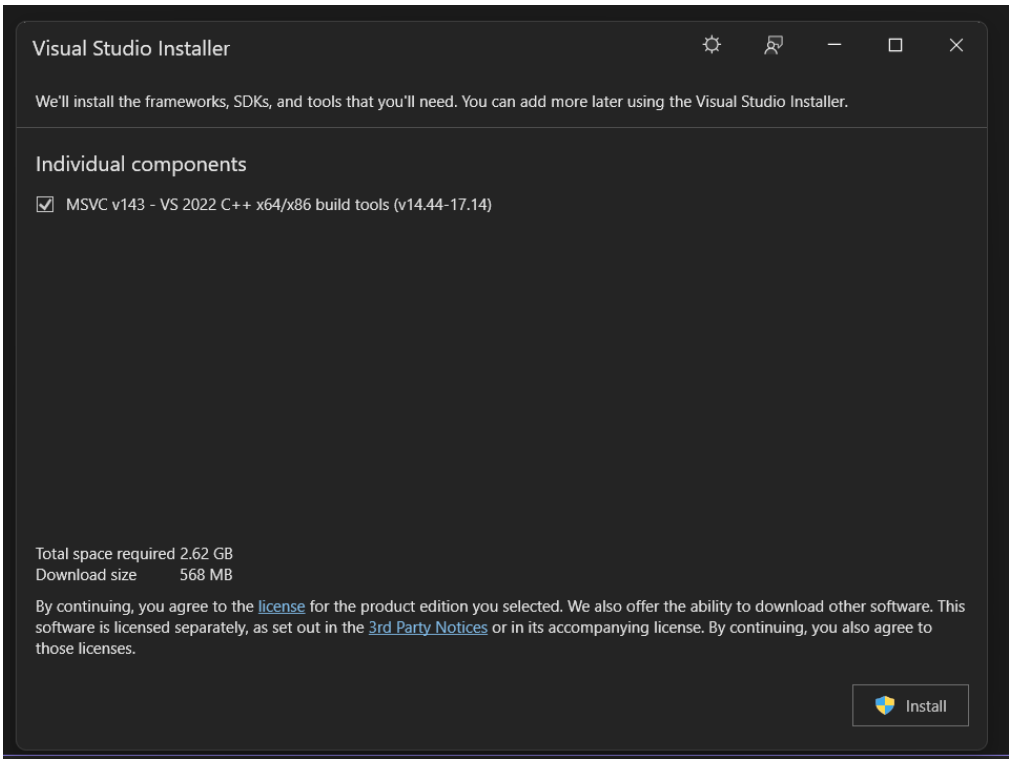


Give a name on your project and then on the location, place it anywhere on your computer (where is most convenient for you, so you can open the project easily). On top, select Visual Studio 2022. Although we just downloaded VS2026, it should still work without any problems. Then press okay. This will open a new VS project. Once the project is open, the editor will inform you of some missing dependencies. To resolve that, press Install all in the top right (as can be seen in the figure below), and then press Apply.

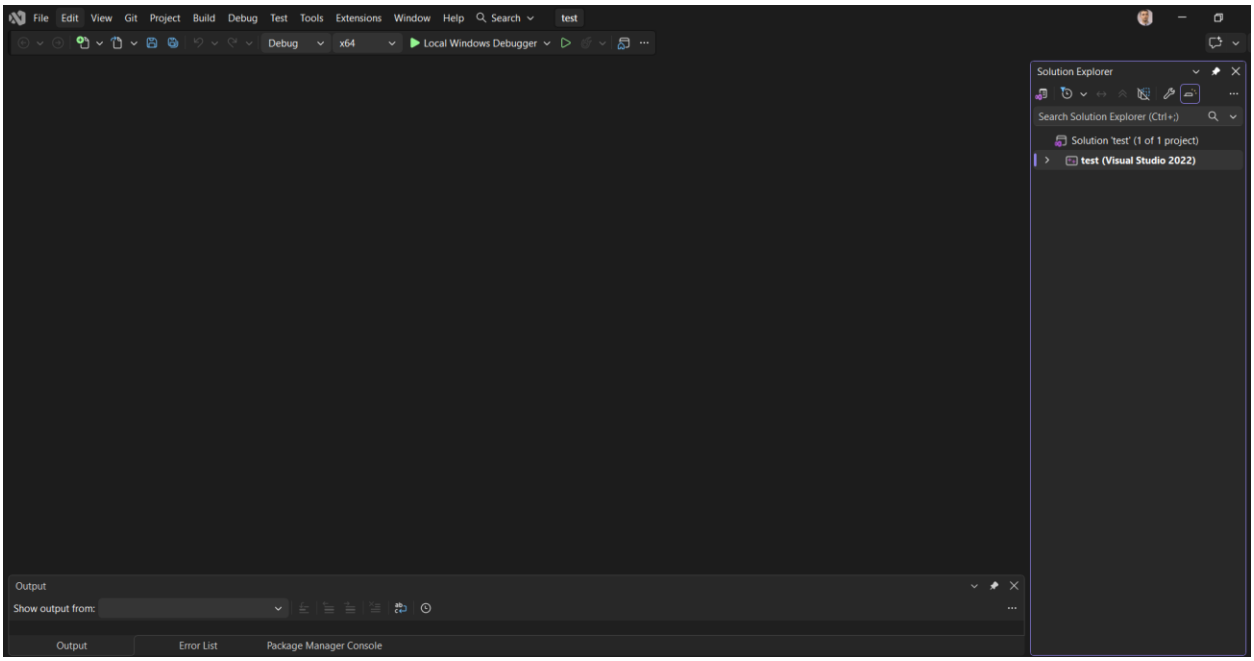
If you receive an error message and the editor is not opening, then delete the folder you just created and repeat the same process.



Once you press apply, the editor will ask you to install the missing dependencies. Press install and wait until the installation is completed. Once this is completed, restart your computer.

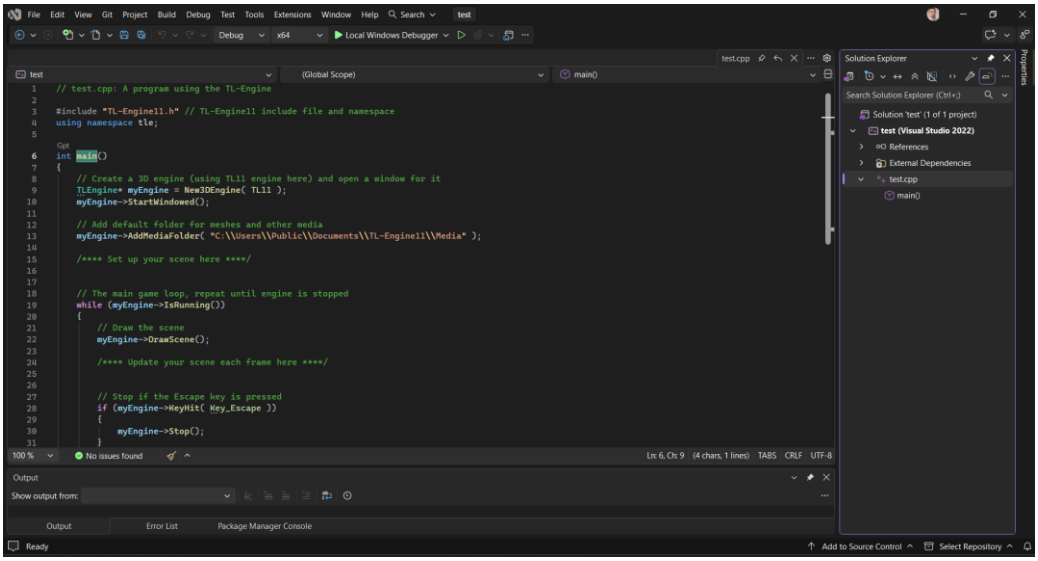


Once your computer is open, in the navigation bar search again for Create new TL11 project and create a new project. If you completed all the previous steps, then VS should open and look like the figure below.



In the above figure, you can see the Solution Explorer. If this is missing, then on the top navigation bar, press on View and select the “Solution Explorer”.

Subsequently, click on the arrow next to the project in the Solution Explorer. This will expand the project. At the bottom, you can find a .cpp file. This is the file that holds the code for your project. Click on that file. This will open the cpp file in front of you. The code you see is the code that a TL project needs to create and run a game project.



Lastly, to test if everything is installed properly, press the “Local Windows Debugger” on top of the file. This will run your code and subsequently open a console window and an empty window, which is the window that your game will run in.

Once you reach this point, you are ready to start with the lab activities.

Happy Learning!